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Effect of Wall Heat Flux on Fluid Flow and Thermal Characteristics of a Turbulent Dual Jet

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ABSTRACT

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Keywords:

offset dual jet, fluid flow and thermal characteristics, wall heat flux

In this study, the two-dimensional, incompressible numerical analysis is carried out to understand the turbulent fluid flow physics and thermal characteristics of a dual jet consisting of a wall jet and a parallel offset jet using standard $k-\epsilon$ RNG turbulence model. The study is carried out to compare the flow and thermal features of the dual jet for an offset ratio of 3, 7 and 9 at different Reynolds numbers in the range of 5,000 to 20,000. The results of the flow field and turbulent behavior of a dual jet are illustrated in the form of temperature contours, velocity vectors, streamlines, and turbulent kinetic energy at different Reynolds number and offset ratio. The effect of offset ratio and Reynolds number on recirculation region and intensity of upper and lower vortices are discussed in detail. The velocity variation along the axial and vertical direction are also discussed.

1. INTRODUCTION

The turbulent flow is playing a vital role in modern days' research and academic applications. It has various applications in industries as well as the devices which contains flow deflection, cooling of gas turbines, combustion chamber, an automobile demister, mixing process in boiler combustion chambers, heat exchangers, engine carburetor systems, aircrafts ventilation, air-conditioning systems, etc. [1-3]. The turbulent jet is mainly classified into a free jet, wall jet, offset jet, dual jet which depends on the placement of jet. A dual jet containing a plane wall jet at the base and a jet with a distance have additional advantages than other jets. If the jet is placed horizontally at the wall is known as wall jet. Also, a jet is placed with some distance from the base is known as offset jet. The wall and offset jet combinedly known as a dual jet.

Rathore and Das [4] conducted a comparative numerical study of flow field analysis and turbulent behavior of wall jet using a low-Reynolds number $k-\epsilon$ model proposed by Launder and Sharma with a standard $k-\epsilon$ model. It was noticed that the Moffatt vortices were occurred in the secondary recirculation region near the wall using a low-Reynolds number modeling approach.

Kumar and das [5] conducted a two-dimensional numerical analysis to compare the momentum flux along the axial direction of a wall jet, dual jet, and offset jet. It is noticed that a similarity profile occurred along the downstream direction. The variation of Reynold stress, pressure variation, turbulence kinetic energy, etc. was compared.

Rao and Das [6] studied the heat transfer characteristics of a wall and offset jet by varying the offset ratio. The constant wall temperature and heat flux boundary condition along with Reynolds number (10000-40000 at Prandtl number of 0.71) were considered in this study.

Mondal et al. [7-9] numerically investigated the interaction between wall jet and offset jet at different velocity ratio (0.78-1.34) for Reynolds number 10000. The height of the wall jet was varied in the range of 0.2 to 2. It is noticed that two counter-rotating vortices occurred for jet height 0.2 whereas a periodic vortex has occurred for higher jet height (greater than 0.3).

Singh et al. [10] studied the heat transfer and fluid flow phenomenon of a dual jet striking on the wavy surface. The amplitude of the wavy surface was varied between 0.1 - 0.7 and has 10 number of cycles. The offset ratio was fixed to be 7. It is observed that an instantaneous pressure drop is obtained in the recirculation zone for the wavy surface.

From the literature review, it is observed that the effect of bottom wall heat flux on flow patterns and thermal behavior of a turbulent dual jet need to be investigated. This work fulfills this literature gap.

In this paper, the effect of wall heat flux on the mixing of fluid of a dual jet consisting of plane wall and jet and parallel offset jet are studied. Dual jet will behave like a wall jet after the combined region where the inlet jets meet in the flow region. The effect of the offset ratio under different Reynolds number is also studied which have a significant effect on the mixing process. For different types of flow domains, temperature distribution will change because of changes in flow characteristics by different parameters like geometry, inlet fluid temperatures. The present results are validated with the experimental results available in the literature.

The paper is systematized as follows: Section 2 discusses the computational domain, boundary conditions, governing equations, numerical issues, and model validation. Section 3 describes the important findings of the study. Section 4 concludes the results.



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A Comprehensive Review on CNTs and CNT-Reinforced Composites: Syntheses, Characteristics and Applications

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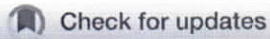
Abstract

Since their first synthesis, carbon nanotubes (CNTs) gained remarkable research interest owing to their astonishing mechanical properties and extensive range of potential applications in various sectors, such as aerospace, automobile, biomedical, defence, energy, etc. This paper covers numerous characterization techniques, synthesis approaches and properties of the CNTs, reported by earlier researchers in the past. The technological and industrial needs for the development of lightweight nanocomposites have led to the significant advancements in the preparation of CNT-reinforced composites. In preliminary sections, the properties and applications of the CNT-reinforced nanocomposites are elaborated along with the issues related to their preparation. Here, various nanotubes synthesis processes, such as arc discharge, laser ablation and chemical vapour deposition, are exemplified with the support of published works. Furthermore, we have also addressed the several surface modification techniques of CNTs, such as purification, functionalization and dispersion, which make this review novel and exhaustive. In order to address the limitations and challenges incurred during the preparation of various CNT-reinforced polymer/metal matrix composites, an extensive collection of the published literature is reported and discussed, thoroughly. Based on this exhaustive review, some specific observations are made which would facilitate upcoming researches to explore the research opportunities in the preparation of CNTs and CNT-reinforced composites and their potential applications for the high-performance structures/components.

RESEARCH ARTICLE | JULY 16 2020

A novel design of an alternator tester for automobiles

Vikrant Tapas; Rituraj Chandrakar; Prem Shankar Sahu; Anil Kumar;
Haldhar Ram Chandrakar



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The alternator is the most essential equipment in an automobile to charge the battery. A modern vehicle is fitted with new accessories which run on the battery. Hence the health of battery is very important in new vehicles. This research paper gives a detailed study of maintenance of alternator and also suggests an alternator tester which is handy and very useful equipment for Alternator testing.

Topics

Alternator, Batteries, Educational assessment

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PAPER

Influence of TiO₂ and MWCNT nanoparticles dispersion on microstructure and mechanical properties of Al6061 matrix hybrid nanocomposites

Sourabh Kumar Soni¹  and Benedict Thomas¹ 

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STUDY OF CAUSES AND REMEDIES OF BULLWHIP EFFECT IN INDUSTRY

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Abstract - Bullwhip impact is a phenomena of increment in variety of interest as one problem of the gracefully in supply chain. Bullwhip impact affects the maintainable development of the ventures. Likewise it might reason for shortcoming of flexibly chain of any industry. In this report the causes & effects of bullwhip effect is determined from Cement Industry. In this report remedies for counter the Bullwhip effect is also given. It can reduce by using ISM technique.

Key Words: Bullwhip Effect, Supply Chain Management, Inventory Management, Variability, ISM

1. INTRODUCTION

The bullwhip impact is one of the most famous and praised ideas in the activities the executives/research field. The term 'bull-whip' was authored to portray the impact by which sluggish buyer request makes enormous swings underway for the providers at the opposite finish of the flexibly chain. This is practically equivalent to the handle of the bullwhip causing an uproarious split at the popper.

The bullwhip impact is now and then alluded to as 'request intensification', 'fluctuation enhancement' or the 'Forrester impact'. This impact becomes critical when the expense from changes underway/requesting exceeds the expense of holding stock. Throughout the long term, proof has recommended that bull-whip costs assume a critical job in certain organizations. Bullwhip expenses can be related with setting up and closing down machines, sitting and extra time in the remaining task at hand, employing and terminating of the workforce, exorbitant upstream stock, trouble in measuring and planning, frameworks anxiety, and helpless provider/client connections, among different results.

2. LITERATURE REVIEW

LIU Hong et al. & WANG et al. Ping the Bullwhip effect is not a newly originated topic but it is an age old phenomenon. In the early 1960s Professor Forrester studied and illustrated the existence of Bullwhip effect by a series of case studies. Professor Sterman of MIT gave his theory on the existence of the Bullwhip effect, in an experiment of "Beer Distribution Game".

Chen et al. Simchi-Levi et al and Shen found that, independent local optimization choices of each manager without global vision, was one of the cause to originate Bullwhip effect. This is one of the most important factor causing Bullwhip effect. Bolton and Katok studied and

analyzed the relationship between the inventory conditions and the inventory or stocks.

Lee et.al (1997) introduced the concept of BWE with an analytical frame. In Lee et.al (1997b), the implications of BWE to managers was elaborated. In an endeavor to key out the substantial grounds of bullwhip effect, this observed phenomenon comprises of two main categories, viz behavioral grounds and operational grounds. The behavioral grounds have chiefly been examined through case studies or tested in laboratory setting. A simulation instrument such as Beer game has been used in laboratory setting for empirical analysis of behavioral grounds. The bullwhip's operational grounds encompass the sixteen main causes, viz. Demand forecasting, order batching, price fluctuation, rationing and shortage gaming

3. CAUSES OF BULLWHIP EFFECT

1. Demand Forecast Updating: Request gauge refreshing is a consequence of the view of chiefs in the chain. Chiefs make a psychological model of the flexibly chain working and the interest designs they watch. This psychological picture impacts their dynamic. At the point when each downstream part submits a request, the upstream part straightens out the interest gauge and afterward puts in a request to the upstream accomplice in the gracefully chain.

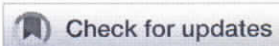
2. Order Batching: In a gracefully chain, each flexibly chain part puts orders on an upstream part utilizing some stock renewal instrument. As request exhausts stock, an organization or a flexibly chain element may not structure persistently, however rather will amass stock recharging necessities from its provider. The distributor/retailer doesn't submit a request with the upstream part when he gets data about the exhausting amount of item. He follows his own style of request setting. He may arrange week by week or month to month, rather than requesting oftentimes as required to diminish the expense per request.

3. Price Fluctuation: Wholesalers occasionally have different plans and advancements like discounts and coupons to build client interest for the item. The purchasing behavior of the client during such periods doesn't reflect purchasing needs, however is an occasional condition. This variety in purchasing behavior is a lot higher than the variety in the utilization rate. Advancements can likewise be an impetus for purchasing more than the interest necessities. These advancements bring about some type of value changes, which power the client to purchase more than required or to sit tight at an ideal cost before reordering or purchasing. This yields brief advantages for one part in the flexibly chain, however makes the Bullwhip Effect and expanded costs upstream.

RESEARCH ARTICLE | JULY 16 2020

To analyze the different popular design of low-pressure acetylene gas welding generators in India

Vikrant Tapas; Rituraj Chandrakar; Prem Shankar Sahu; Anil Kumar;
Haldhar Ram Chandrakar



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The low-pressure gas welding is very popular in sheet welding. The small automobile garages, body repairing workshops used this type of welding extensively in India. This research paper focused on the study of the construction and operation of popular acetylene gas generators models used in low-pressure welding. After studying on different parameters the authors found that prefabricated or readymade generators are found best for safe operational uses after comparing them on different parameters with fabricated generators. The authors firmly believe that the existing fabricated acetylene generators must be changed with a prefabricated model to improve the quality of the weld and safe welding practices.

Topics

Welding

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Comparison Analysis and Implementation of Prediction of Heart Disease

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Abstract

In the current decade Heart disease is most common for all from 14-60 years of age. Now a days heart disease having many types of constraint to predict the information like hyper tension, Blood pressure increase, Blockage of Nervous system. So that the percentage of blockage must identified which surgery is good for the health of patient. In this paper we are trying to find the better method to predict and we also used algorithms for prediction. We use various algorithm for the prediction of blockage of nerves like Naïve Bayes, algorithm is analyzed on dataset based on hazard factors. Here we also try to use decision trees and grouping of algorithms for the calculation of heart disease based on the features. The results shown the comparison of various four algorithms of data mining which compare their accuracy of the result which one is better through graph..

Keywords: *Decision tree, Data mining, Heart Disease Prediction, Naïve Bayes, K-means, Machine learning.*

I. INTRODUCTION

The main topic is prediction using machine learning technics. Machine learning is widely used now a days in many business applications like e commerce and many more. Prediction is one of area where this machine learning used, our topic is about prediction of heart disease by processing patient's dataset and a data of patients to whom we need to predict the chance of occurrence of a heart disease. Data mining practices that are valuable in heart disease forecast with the assistance of dissimilar data mining tools that are accessible. If the heart doesn't function properly. The major challenge that the Healthcare industry faces now-a-days is superiority of facility. Diagnosing the disease correctly & providing effective treatment to patients will define the quality of service. Even though heart disease is acknowledged as the supreme chronic sort of disease in the world, it can be most avoidable one also at the same time. A healthy way of life (main prevention) and timely analysis. Heart expert's create a good and huge record of patient's database and store them. There is always a need to find out the Heart data set, such as Blood clotting, nerves damage, how much percentage blocking must be the challenging part of people. Additionally, there are situations when silent heart attack is preferred: for example, during an operation, seivour attack must be held in some times. It is hard for most people who are not familiar with a heart failure will take up certain parameters without an interpreter.

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Friend Recommendation on Social Media

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Abstract – Friend recommendation is one of the most popular characteristics of social network platforms, which recommends similar or familiar people to users. The concept of friend recommendation originates from social networks such as Twitter and Facebook, which uses friends-of-friends method to recommend people. We can say users do not make friends from random people but end up making friends with their friends' friends. The existing methods have narrow scope of recommendation and are less efficient. We put forward a new friend recommendation model to overpower the defects of existing system. For better friend recommendation system with high accuracy, we will use collaborative filtering method to compare similar, dissimilar data of users and will make a recommendation system which gives user to user recommendation based on their similar choices, activities and preferences. Location based friend recommendation system are becoming popular because it brings physical world to digital platform and gives better insight of user's preferences or interest. This recommendation system will increase the scope of recommendation from one user to other with similar set of interest and their location.

Keywords– Friend recommendation, collaborative filtering, social network, Recommendation system.

I. INTRODUCTION

Recommendation is one of the most common and fundamental service in LBSN platform which recommends familiar or interested user to each other. About 71% of internet users were online social network users and they will grow in near future. Social networking is very popular online activities with high rate of user interactions & expanding mobile possibilities. The growth rate in use of smart phones and mobile devices is very rapid and has opened up new areas of mobile social networks with increased features. With over billions of monthly active users on social network. Facebook is currently the market leader in terms of user engagement reach and scope [1].

Recent advances in localization techniques have improved social networking services, allowing users to share their locations and location-related contents. Such type of social networks is referred as location-based social networks (LBSNs). LBSNs is equipped with type of friend recommendation which utilizes user's historical location information.

Traditional friend recommender engines provide a user with promising candidates to make friends based on their profiles, social structure & interactions. Location information can improve the effectiveness of recommendations. The basic idea is that user location histories reveal choices, and thus users with similar

location histories have similar choices & have higher probability to be friends [2].

Friend recommendation service is used for conventional social networks. But there are very less algorithms that exploit LBSN data in recommendation. Earlier methods generally use GPS information to find the resemblance between users. When compared to GPS information, check_in information gives more context depended information. Furthermore, most of the LBSNs collect check_in information than the GPS trajectory data. The objective of our proposed recommendation systems is to include user profiles, interest, and user location histories (check_in data) and apply collaborative filtering methods for user to user recommendation to increase scope of recommendation and make it more efficient [3].

A location-based social network doesn't mean concatenating a location to an existing social network to allow people to share location related information and activities, but LBSN is also made up of the new social structure of individuals connected together by the inter dependency of their locations in the real world & their location-tagged media like text, image and video [4]. Physical location does not only include the instant point location of an individual at a given timestamp but the location history of an individual over a specified time period. Also, the knowledge, common interests, and preferred activities are derived from an individual's location information and location related content affects the social relations in LBSN [5].

A Review of Friend Recommendation on Social Media

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Abstract

Friend recommendation is one of the most popular characteristics of social network platforms, which recommends similar or familiar people to users. The concept of friend recommendation originates from social networks such as Twitter and Facebook, which uses friends-of-friends method to recommend people. We can say users do not make friends from random people but end up making friends with their friends' friends. The existing methods have narrow scope of recommendation and are less efficient. We propose a new friend recommendation model to overcome the shortcomings of existing system. For better friend recommendation system with high accuracy, we will use collaborative filtering method to compare similar, dissimilar data of users and will make a recommendation system which gives user to user recommendation based on their similar choices, activities and preferences. Location based friend recommendation system are becoming popular because it brings physical world to digital platform and gives better insight of user's preferences or interest. This recommendation system will increase the scope of recommendation from one user to other with similar set of interest and their location.

Keywords: Friend Recommendation, Collaborative Filtering, Social Network, Recommendation System.

I. INTRODUCTION

Friend recommendation is one of the most common and fundamental service in LSBN platform which recommends familiar or interested user to each other. By 2017, 71% of internet users were social network users and they are expected to grow in near future. Social networking is the most popular online activities with high user engagement rate and expanding mobile possibilities. The rapid growth in use of smart phones and mobile devices has opened up the possibilities of mobile social networks with increased features. With over 1.86 billion monthly active users, social network Face book is currently the market leader in terms of user engagement reach and scope. Recent advances in localization techniques have improved social networking services, allowing users to share their locations and location-related contents. Such type of social

recommendation systems is to include user profiles, interest, and user location histories and apply collaborative filtering methods for user to user recommendation to increases scope of recommendation and make it more efficient.

1. Recommender Systems:

A Recommender system is an information filtering system that predicts preference a user would give to something. With more use of e-commerce sites, it has become very easy for the users to find the items of their interest without wasting a lot of time. Websites like Amazon and E bay examples for Recommender systems which provide recommendations to the users based on their search history and purchase history. Recommender systems provide recommendations of almost all the items ranging from books to movies to music. Face Book and Twitter are also recommender sites which